

TAN ZHENG XUN

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EDUCATION

Digipen Institute Of Technology Singapore

Sep 2022 - Present

- **Bachelor of Science in Computer Science in Interactive Media and Game Development**
- GPA: 4.11/5.0
- Relevant Coursework: Data Structures & Algorithms, Operating Systems
- Expected Graduation: May 2026
- **Hackathon: CoDE 2023 Runner Up**

EXPERIENCE

Venti Technologies

Apr 2025 - Present

Simulation Engineer Intern

- Designed and maintained a large-scale autonomous-vehicle simulation framework supporting complex multi-agent scenarios with strict structural and execution constraints.
- Developed tools to validate structured simulation inputs (YAML/JSON), preventing malformed configurations and unsafe state transitions.
- Built automation scripts to generate, audit, and refactor simulation scenarios at scale, improving consistency and reducing manual errors.
- Analyzed logs and performance metrics to detect abnormal execution patterns and resource spikes, supporting system reliability and regression detection.
- Worked extensively in Linux environments, applying defensive programming practices in Python and C++.

PROJECTS

IRON ENGINE | Custom 3D Game Engine, Team of 12

Jan 2025 - April 2025

Systems Programmer

- Developed core engine systems utilizing external libraries such as ENTITAS for entity management.
- Designed a behaviour tree engine for designers to simplify scripting in game creation.
- Utilized run-time type reflection (RTTR) as a channel for communication of components, allowing for objects to inspect itself.

INSIGHT ENGINE | Custom 2D Game Engine, Team of 6

Apr 2024 - Sep 2024

Technical Lead | Programmer

- Led the development of core engine systems, including architecture design, ECS, asset management, multi-threading, memory management, and scripting integration Mono C#.
- Built a high-performance ECS handling 1M+ entities with minimal overhead.
- Implemented a multi-threaded asset loader, reducing load times by 76%.
- Developed a custom memory manager, improving runtime performance by 32%.
- Developed core engine features such as prefabs, input, controller and audio support, messenger system.

SKILLS & ACHIEVEMENTS

Language: C, C++, C#, Python, Kotlin

Tools & Frameworks: Unity, Mono C#, Git, Linux(Ubuntu)

System & Design patterns: Memory management, multi-threading, ECS, performance profiling